

Windows Profile Error Recovery

Issue: "User Profile Service service failed the sign-in" or user is logged into a TEMP profile.
TEMP.DOMAIN.001, .002 folders accumulating in C:\Users.

Decision Tree

```
User reports profile error at login
■■■ Getting a TEMP profile (desktop empty, settings missing)?
■   ■■■ YES → Check C:\Users for TEMP.* folders
■   ■   ■■■ Multiple TEMP folders → FIX A (SID .bak collision)
■   ■   ■■■ One TEMP folder → Does original profile folder exist?
■   ■       ■■■ YES → FIX A
■   ■       ■■■ NO → Profile deleted/corrupted → FIX B
■   ■■■ "User Profile Service failed the sign-in" – can't log in at all
■       ■■■ Can log in as local admin? → FIX A from admin context
■       ■■■ Cannot log in at all → Boot Safe Mode → FIX A
■■■ Happens to ALL domain users on this machine?
■   ■■■ YES → Machine-level issue: check disk space, Default profile integrity
■   ■■■ NO → User-specific SID issue → FIX A
```

FIX A: SID .bak Registry Repair

Fixes 90%+ of TEMP profile cases.

1. Log in as local Administrator.
2. Open **regedit** as Administrator.
3. Navigate to **HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList**.
4. Find the affected user's SID. You'll typically see:

```
S-1-5-21-xxx-xxx-xxx-xxxx    ← Points to TEMP path
S-1-5-21-xxx-xxx-xxx-xxxx.bak ← Points to real profile path
```

5. The **.bak** entry is the original. The non-.bak is the failed redirect.
6. Rename the non-.bak entry → add **.old** suffix.
7. Rename the .bak entry → remove **.bak** suffix.
8. In the renamed entry, set both **RefCount** and **State** to **0**. See the note below

if either value is missing.

9. Close regedit.
10. Delete **C:\Users\TEMP.*** folders (user must be logged off).
11. Reboot.

If RefCount or State is not present

These are **REG_DWORD** values. On many profiles - especially ones that never fully

loaded - **RefCount** (and sometimes **State**) simply does not exist yet. That is normal.

Do NOT skip the step; CREATE the missing value and set it to **0**, because Windows reads

a missing value differently from an explicit 0.

- **RefCount** = the count of active loads of the hive. Missing usually means 0 loads, but creating it as 0 makes that explicit so Windows does not treat the profile as in-use and re-route to a TEMP profile.

- **State** = the profile state flags. 0 means "normal / healthy." A non-zero value (commonly 0x8000 = 32768, or other bits) marks it corrupt/temporary; forcing 0 clears that flag.

Create a missing value in regedit: with the renamed (non-.bak) SID key selected, right-click in the right pane > New > DWORD (32-bit) Value > name it exactly **RefCount** (or **State**) > double-click it > set Value data to 0 (Base: Hexadecimal or Decimal, 0 is the same either way).

Or set/create both from the command line (replace the SID with the real one from step 4; the key must already be the renamed non-.bak key):

CMD:

```
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-xxx-xxx-xxx-xxxx" /v  
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-xxx-xxx-xxx-xxxx" /v
```

PowerShell:

```
$sid = "S-1-5-21-xxx-xxx-xxx-xxxx"  
$key = "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\$sid"  
New-ItemProperty -Path $key -Name RefCount -PropertyType DWord -Value 0 -Force  
New-ItemProperty -Path $key -Name State -PropertyType DWord -Value 0 -Force
```

New-ItemProperty ... -Force and **reg add ... /f** both CREATE the value if absent and overwrite it if present, so the same command works either way.

FIX B: Profile Folder Missing

1. Delete the orphaned SID entry from ProfileList (back it up first).
2. Delete any TEMP folders for that user.
3. Reboot — Windows creates a fresh profile on next login.
4. Restore documents from backup if available.

FIX C: Disk Full Causing Profile Failures

Profile creation fails silently when the drive is full.

```
# Check free space  
(Get-PSDrive C).Free / 1GB
```

CMD:

```
dir C:\ | find "bytes free"

# If under 2 GB, free space first
.\Invoke-HelpDeskOrchestrator.ps1 -RelieveDiskPressure -CleanupTemp -DisableHibernation
```

Then proceed with FIX A.

Bulk Cleanup

```
# List TEMP profile folders
Get-ChildItem C:\Users -Directory | Where-Object { $_.Name -match "^TEMP\." }
```

CMD (list TEMP profile folders):

```
dir /ad "C:\Users\TEMP*"

# Remove them (ensure users are logged off)
Get-ChildItem C:\Users -Directory | Where-Object { $_.Name -match "^TEMP\." } |
    Remove-Item -Recurse -Force -ErrorAction SilentlyContinue
```

Prevention

- Run **-AuditProfileHealth** weekly to catch .bak entries early.
- Maintain minimum 10 GB free on system drives.
- If recurring across multiple machines, investigate domain trust, GPO, or SID replication issues.

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